OBSCYRATECH

CYBERSECURITY • TRAINING • SERVICES

PROFESSIONAL SYLLABUS

Certification Program

Investigate cyber incidents through packet captures, logs, intrusion detection, and network monitoring. Learn how attackers communicate, reconstruct incidents from network evidence, and produce forensic findings that support incident response and legal cases.

MODULES	LEVEL	FORMAT	OUTCOME
8 Modules	**Intermediate**	**Labs + Capstone**	**DFIR / SOC Ready**

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Mysore, Karnataka, India

Course Description

Network Forensics focuses on the identification, collection, preservation, analysis, and reporting of network-based evidence. The course teaches students how to investigate cyber incidents using **packet captures, logs, intrusion detection systems,** and **network monitoring tools**.

Students will learn how attackers communicate across networks, how to reconstruct incidents from network evidence, and how to produce **forensic findings that support incident response and legal investigations**.

PREREQUISITES

- ✓ Networking Fundamentals
- ✓ Cybersecurity Fundamentals
- ✓ Basic TCP/IP Knowledge
- ✓ Basic Linux Knowledge (Recommended)

Learning Objectives

- ✓ Understand network forensic investigation methodologies.
- ✓ Capture and analyze network traffic.
- ✓ Investigate network-based attacks and incidents.
- ✓ Examine logs and network artifacts.
- ✓ Detect malicious communications and indicators of compromise.
- ✓ Analyze encrypted traffic and cloud-based evidence.
- ✓ Utilize industry-standard network forensic tools.
- ✓ Produce professional forensic reports.

Syllabus Index

CORE CURRICULUM

Course Modules

MODULE 01 Introduction to Network Forensics

LEARNING OBJECTIVES

- Understand the purpose of network forensics.
- Learn the forensic investigation lifecycle.
- Understand digital evidence concepts.

KEY TOPICS

- What is Network Forensics?
- Network Evidence Sources
- Investigation Lifecycle
- Types of Cyber Incidents
- Chain of Custody Basics

PRACTICAL ACTIVITY Analyze a real-world network breach case study.

MODULE 02 Network Fundamentals for Investigators

LEARNING OBJECTIVES

- Understand network communication.
- Interpret packet structures.
- Learn protocol fundamentals.

KEY TOPICS

- TCP/IP Model
- OSI Model Review
- IP Addressing
- Ports and Protocols
- Packet Structure

PRACTICAL ACTIVITY Inspect packets using Wireshark.

MODULE 03 Traffic Capture & Evidence Collection

LEARNING OBJECTIVES

- Learn packet acquisition techniques.
- Understand evidence preservation.
- Validate captured data.

KEY TOPICS

- Packet Capturing
- PCAP Files
- SPAN Ports
- Network TAPs
- Evidence Preservation

PRACTICAL ACTIVITY Capture and save network traffic using Wireshark and tcpdump.

MODULE 04 Protocol Analysis & Traffic Investigation

LEARNING OBJECTIVES

- Analyze common protocols.
- Identify suspicious communications.
- Understand protocol behavior.

KEY TOPICS

- HTTP Analysis
- DNS Analysis
- FTP Analysis
- SMTP Analysis
- DHCP Analysis

PRACTICAL ACTIVITY Investigate suspicious DNS traffic.

MODULE 05 Network Log Forensics

LEARNING OBJECTIVES

- Analyze logs for evidence.
- Correlate multiple data sources.
- Build investigation timelines.

KEY TOPICS

- Firewall Logs
- Proxy Logs
- Authentication Logs
- Web Server Logs
- Event Correlation

PRACTICAL ACTIVITY Create an incident timeline using logs.

MODULE 06 Advanced Traffic Analysis

LEARNING OBJECTIVES

- Reconstruct attacker activity.
- Detect data exfiltration.
- Analyze suspicious sessions.

KEY TOPICS

- Session Reconstruction
- File Transfer Analysis
- Data Exfiltration Detection
- Lateral Movement Indicators
- Network Artifact Analysis

PRACTICAL ACTIVITY Recover transferred files from captured traffic.

MODULE 07

Security Monitoring & Threat Hunting

LEARNING OBJECTIVES

- Understand threat hunting concepts.
- Investigate anomalies.
- Monitor network behavior.

KEY TOPICS

- Network Security Monitoring
- Threat Hunting Methodology
- Behavioral Analysis
- Traffic Baselining
- Investigation Workflows

PRACTICAL ACTIVITY Conduct a threat-hunting exercise using Security Onion.

MODULE 08

Reporting, Legal Considerations & Capstone Project

LEARNING OBJECTIVES

- Create professional forensic reports.
- Understand legal requirements.
- Present investigation findings.

KEY TOPICS

- Forensic Documentation
- Report Writing
- Evidence Presentation
- Legal Admissibility
- Expert Testimony Basics

FINAL CAPSTONE PROJECT

Conduct a complete network forensic investigation involving:

- ✓ Packet Capture Analysis
- ✓ Log Analysis
- ✓ IDS Alert Investigation
- ✓ Timeline Reconstruction
- ✓ Incident Report Preparation

Assessment & Grading

Balanced evaluation across quizzes, labs, assignments, midterm, case studies, and a final capstone project.

ASSESSMENT	WEIGHTAGE
Quizzes	10%
Lab Exercises	25%
Practical Assignments	20%
Midterm Investigation	15%
Case Studies	10%
Final Capstone Project	20%
Total	**100%**

Recommended Tools

Industry-standard network forensic and security monitoring tools used throughout the program.

- **Wireshark**
- **tcpdump**
- **NetworkMiner**
- **Zeek (Bro)**
- **Suricata**
- **Snort**
- **Security Onion**
- **ELK Stack**
- **Splunk**
- **VirtualBox**
- **Sample PCAP Repos**
- **MITRE ATT&CK Reference**

Recommended Reading

Industry-respected books to deepen your network forensics knowledge.

Core Textbooks

- Network Forensics: Tracking Hackers Through Cyberspace
- Practical Packet Analysis
- The Practice of Network Security Monitoring

Practice Platforms

- Malware Traffic Analysis exercises
- CyberDefenders Blue Team Labs
- SANS DFIR challenges
- Honeynet Project network challenges

Career Outcomes

Roles you can pursue after completing the Network Forensics program:

Forensics & DFIR Roles

- Network Forensics Analyst
- DFIR Analyst
- Incident Response Analyst
- Cybercrime Investigator

SOC & Operations Roles

- SOC Analyst (Tier 1 / Tier 2)
- Security Monitoring Analyst
- Threat Hunter
- Security Operations Engineer

Recommended advanced learning tracks at Obscyratech:

Advanced DFIR | **Threat Hunting Specialization** | **Incident Response Management** | **Malware Analysis** | **Cloud Forensics** | **SOC Operations**

Program Promise: This Network Forensics program is an excellent progression after **Network Security** and **Digital Forensics**, and a perfect foundation before advanced **DFIR, Threat Hunting,** and **Incident Response** specializations. Every module blends theory, hands-on labs with real PCAP and log datasets, and a capstone investigation.

OBSCYRATECH

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Hunt attackers across the wire.

Join Obscyratech's Network Forensics program and master the tools and techniques used by DFIR analysts, threat hunters, and SOC investigators worldwide — from packet analysis to capstone-grade incident reporting.

GET IN TOUCH

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LOCATION

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PROGRAM

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